

Assignment 8 (Solutions will be uploaded on March 29)

Below 2 questions help you understand the Lecture Note 8 topics.

1. Use the WAGE2.dta dataset to analyze the following regression model.

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{married} + \beta_3 \text{exper} + \beta_4 \text{tenure} + u,$$

where *educ* is endogenous; three instruments: *sibs*, *birthorder* and *meduc*; *married*, *exper* and *tenure* are exogenous.

(a) Test for instrument relevance.

(b) Use 2SLS to estimate the unknown population parameters

(c) Is instrument exogeneity satisfied? Calculate the *J* statistic. Explain your finding.

(d) Use the Hausman test to test for exogeneity. [Hint: Once you get the OLS or IV result, you can use the Stata command “*estimate store any name*” to store for the estimates in Stata]

[Difficulty: 3 out of 5]

2 Consider a simple regression $y_i = \beta_0 + \beta_1 x_i + u_i$, $i = 1, 2, \dots, n$, where $\text{Cov}(x, u) \neq 0$.

(a) Show that asymptotically (i.e., $n \rightarrow \infty$), $\hat{\beta}_{1,OLS} \approx \beta_1 + \text{Corr}(x, u) \times \frac{\sigma_u}{\sigma_x}$

(b) Show that asymptotically (i.e., $n \rightarrow \infty$), $\hat{\beta}_{1,IV} \approx \beta_1 + \frac{\text{Corr}(z, u)}{\text{Corr}(z, x)} \times \frac{\sigma_u}{\sigma_x}$.

(c) Suppose the instrument exogeneity does not hold and *z* is a weak instrument. How does it affect our $\hat{\beta}_{1,IV}$? Do you still prefer $\hat{\beta}_{1,IV}$ to $\hat{\beta}_{1,OLS}$? Explain your answer.

[Note: Suppose *x* and *y* are random variables, then $\text{Corr}(x, y) = \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x)}\sqrt{\text{Var}(y)}}$]

[Difficulty: 4 out of 5]